

Section 3.1 Worksheet

The Determinant of a Matrix

Directions

Show all determinant calculations. Use cofactor signs explicitly when needed. Problems 1–4 are core; Problem 5 is conceptual.

1. Compute

$$\det \begin{bmatrix} 7 & 2 \\ 3 & 5 \end{bmatrix}.$$

2. For

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{bmatrix},$$

find all nine minors and all nine cofactors.

3. Use cofactor expansion to evaluate

$$\det \begin{bmatrix} 4 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

4. Find the determinant of

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 4 & 5 & -2 & 0 \\ 7 & 1 & 6 & 5 \end{bmatrix}.$$

5. Explain geometrically why a 2×2 matrix with determinant zero collapses area.

Section 3.2 Worksheet

Determinants and Elementary Operations

Directions

Record how every row or column operation changes the determinant.

1. Use row operations to find

$$\det \begin{bmatrix} 3 & 1 & 2 \\ 6 & 5 & 1 \\ 0 & 4 & 7 \end{bmatrix}.$$

2. Suppose $\det(A) = -6$. Find the determinant after each operation:

- (a) interchange rows 1 and 2;
- (b) multiply row 3 by -3 ;
- (c) add 5 times row 1 to row 2.

3. Explain why

$$\det \begin{bmatrix} 2 & 5 & 1 \\ 4 & 10 & 2 \\ 3 & 1 & 7 \end{bmatrix} = 0.$$

4. Use a column operation to evaluate

$$\det \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 4 & 12 & 1 \end{bmatrix}.$$

5. A data matrix has one column equal to the sum of two other columns. State what this implies about its determinant and invertibility.

Section 3.3 Worksheet

Properties of Determinants

Directions

Use determinant properties. Do not multiply matrices unless necessary.

Suppose A and B are 3×3 matrices with $\det(A) = 4$ and $\det(B) = -3$.

1. Find $\det(AB)$, $\det(BA)$, $\det(2A)$, and $\det(B^{-1})$.
2. Find $\det(2AB^{-1})$.
3. If $\det(C) = 0$, state what you can conclude about C^{-1} and the homogeneous system $C\mathbf{x} = \mathbf{0}$.
4. A two-dimensional transformation first scales signed area by -2 and then by 5 . Find the signed area scale of the composition and interpret its sign.
5. If $\det(D) = -2$, find $\det((D^T)^3)$.

Section 3.4 Worksheet

Applications of Determinants

Directions

Show determinant setup and calculation. Use exact values.

1. For

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix},$$

find $\text{adj}(A)$ and A^{-1} .

2. Use Cramer's Rule to solve

$$\begin{cases} x + y + z = 9, \\ 2x - y + z = 5, \\ x + 2y - z = 4. \end{cases}$$

3. Find the area of the triangle with vertices $(0, 1)$, $(6, 2)$, and $(2, 7)$.

4. Find an equation of the line through $(1, 4)$ and $(5, 12)$ using a determinant.

5. Find the volume of the tetrahedron with vertices $(0, 0, 0)$, $(4, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$.

6. Find an equation of the plane through $(1, 1, 0)$, $(3, 0, 1)$, and $(0, 2, 2)$.